

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 19

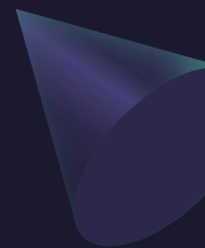


Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 13.1-13.3 (vector functions, their integrals and derivatives, and arc length)
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104



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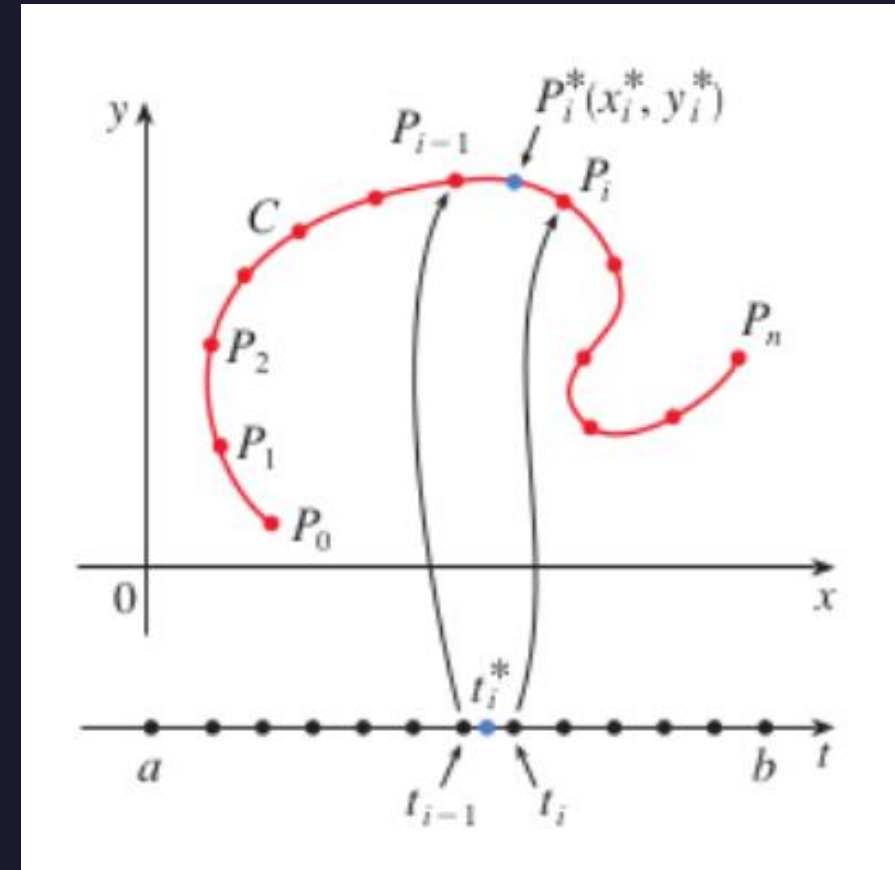
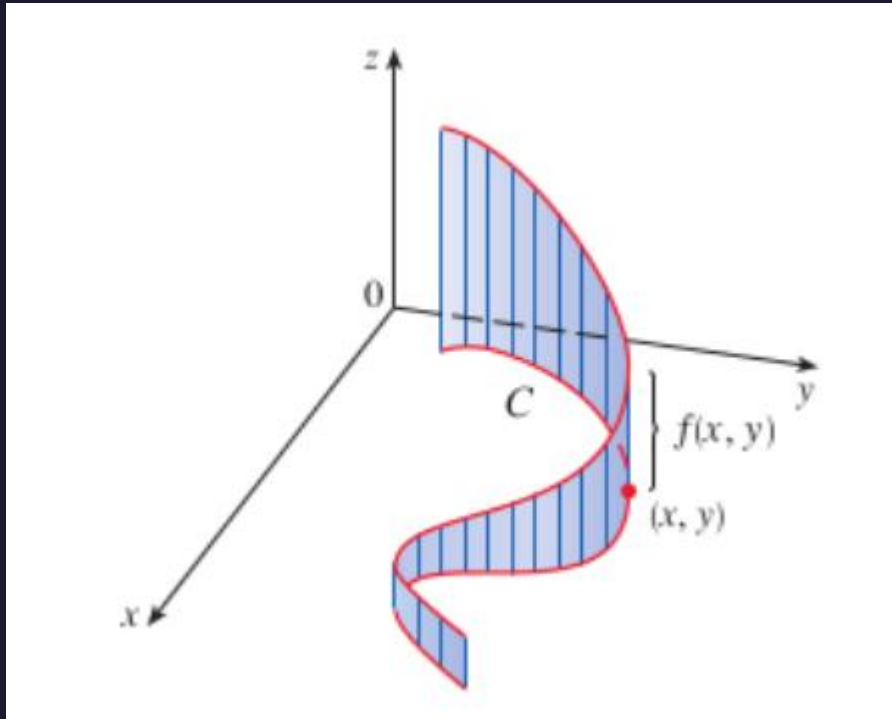
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Line Integrals

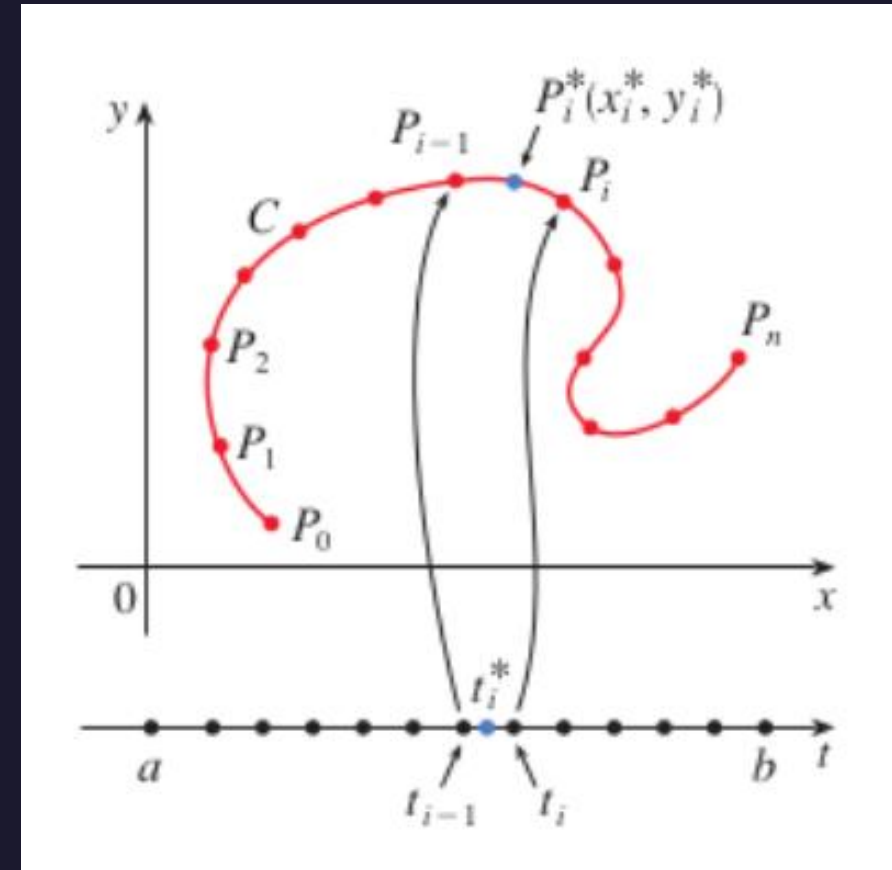
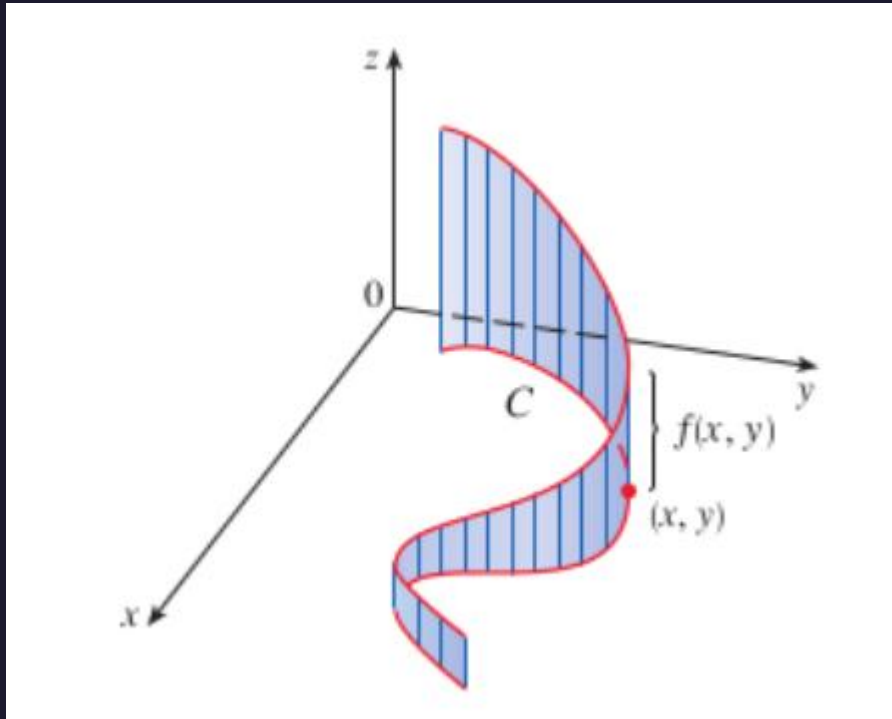


Line Integrals



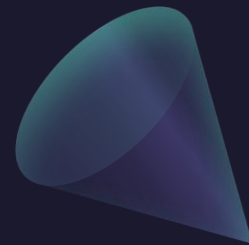
$$\int_C f(x, y) ds = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*) \Delta s_i$$

Line Integrals

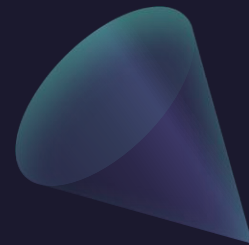


$$\int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

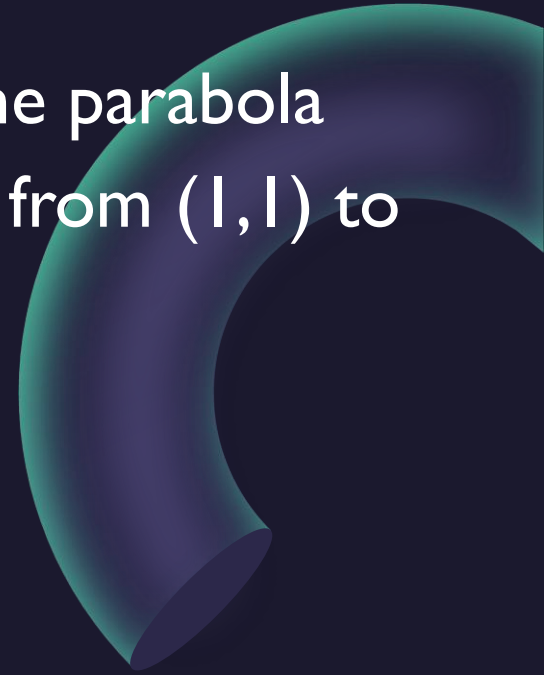
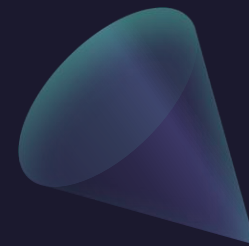
Example: Evaluate $\int_C (2 + x^2 y) ds$ where C is the upper half of the unit circle.



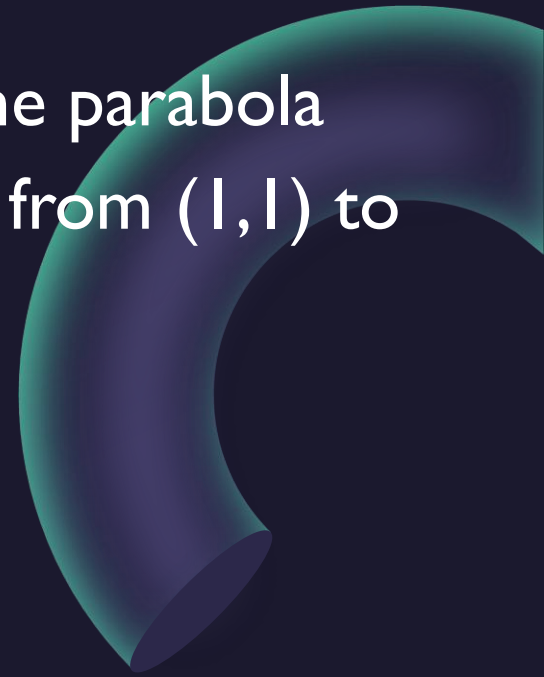
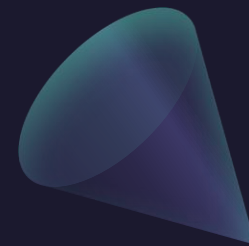
Example: Evaluate $\int_C (2 + x^2 y) ds$ where C is the upper half of the unit circle.



Example: Evaluate $\int_C (2x) \, ds$ where C consists of the arc of the parabola $y = x^2$ from $(0,0)$ to $(1,1)$ followed by a line segment from $(1,1)$ to $(1,2)$.



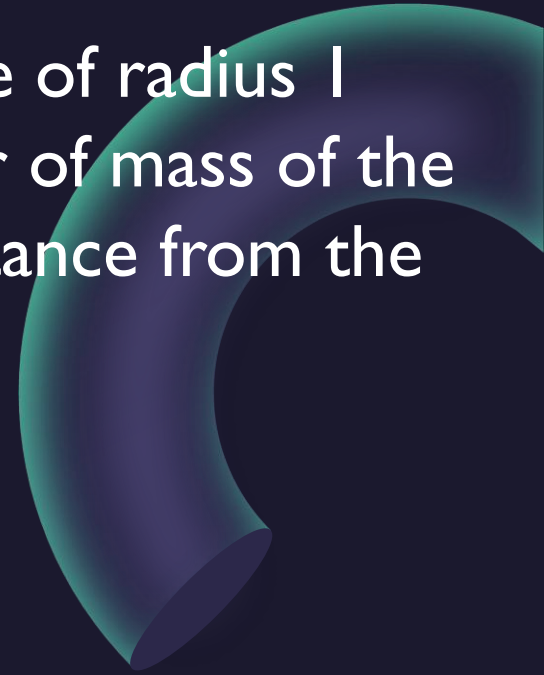
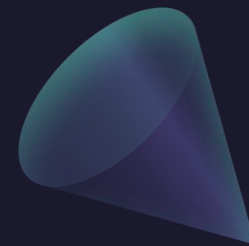
Example: Evaluate $\int_C (2x) \, ds$ where C consists of the arc of the parabola $y = x^2$ from $(0,0)$ to $(1,1)$ followed by a line segment from $(1,1)$ to $(1,2)$.



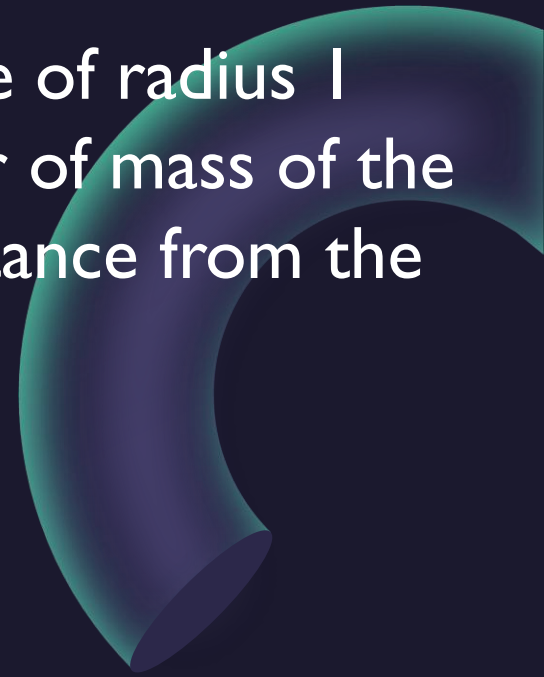
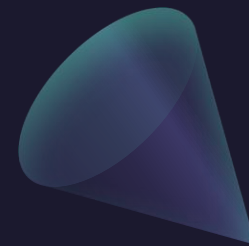
Application of Line Integrals: Center of Mass

$$\bar{x} = \frac{1}{m} \int_C x \rho(x, y) \, ds \qquad \bar{y} = \frac{1}{m} \int_C y \rho(x, y) \, ds$$

Example: A wire takes the shape of the top half of a semi-circle of radius l and it is thicker near its base than near the top. Find the center of mass of the wire if the linear density at any point is proportional to its distance from the line $y = l$.



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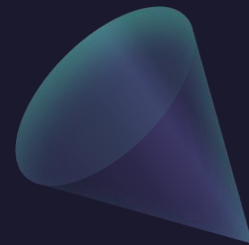


Line integrals with respect to x and y

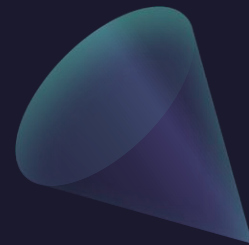
$$\begin{aligned}\int_C f(x, y) dx &= \int_a^b f(x(t), y(t)) x'(t) dt \\ \int_C f(x, y) dy &= \int_a^b f(x(t), y(t)) y'(t) dt\end{aligned}$$

$$\int_C P(x, y) dx + \int_C Q(x, y) dy = \int_C P(x, y) dx + Q(x, y) dy$$

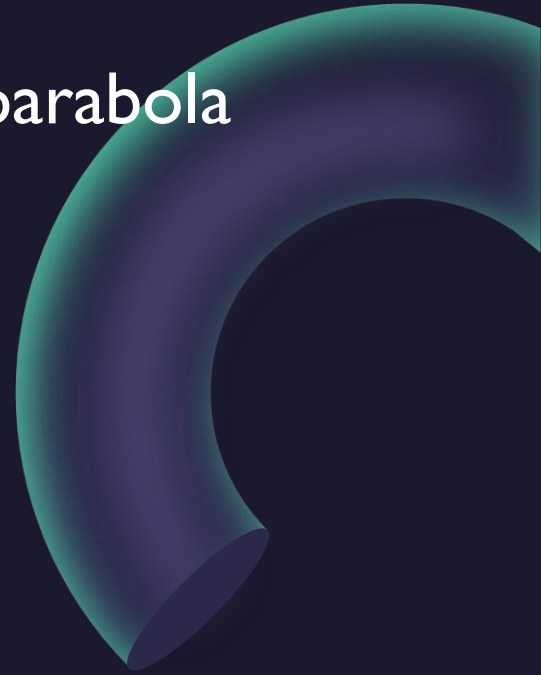
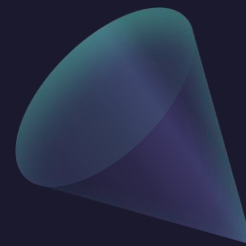
Example: Evaluate $\int_C y^2 dx + x dy$ where C is the line segment from $(-5,-3)$ to $(0,2)$.



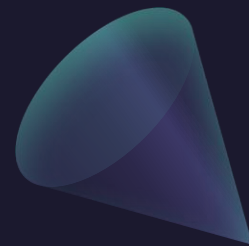
Example: Evaluate $\int_C y^2 dx + x dy$ where C is the line segment from $(-5,-3)$ to $(0,2)$.



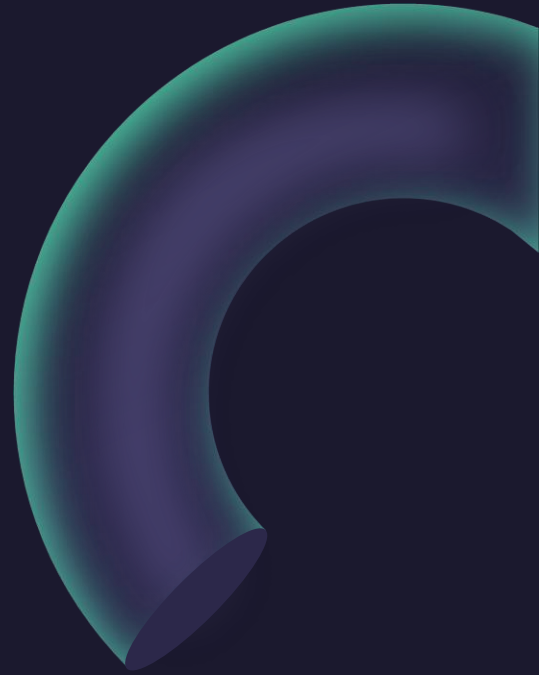
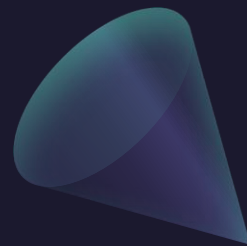
Example: Evaluate $\int_C y^2 dx + x dy$ where C is the arc of the parabola $x = 4 - y^2$ from $(-5, -3)$ to $(0, 2)$.



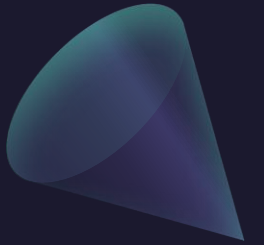
Example: Evaluate $\int_C y^2 dx + x dy$ where C is the line segment from $(-5,-3)$ to $(0,2)$.



Example:

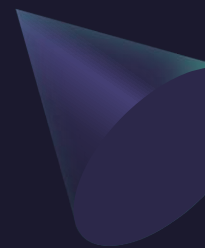


Questions?





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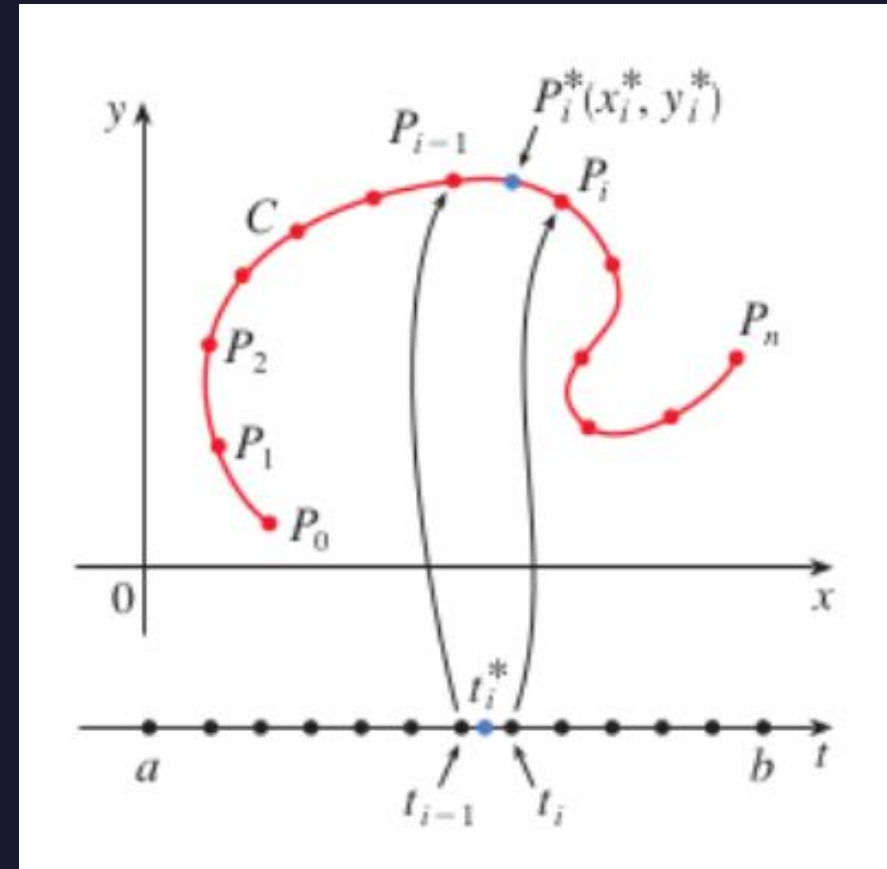
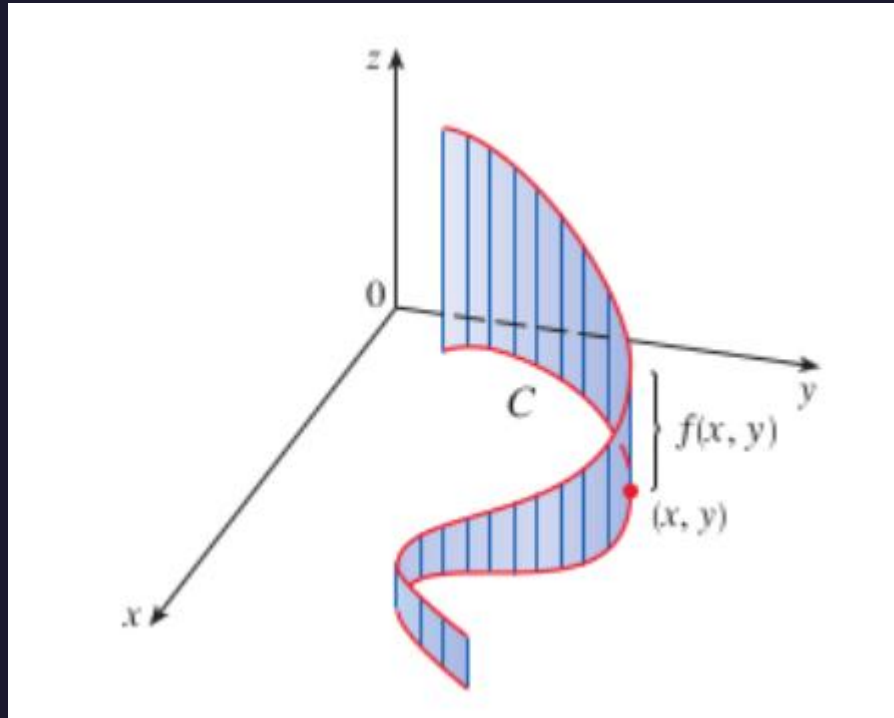
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Line Integrals... in Space



Line Integrals



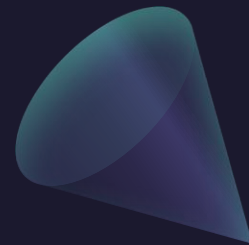
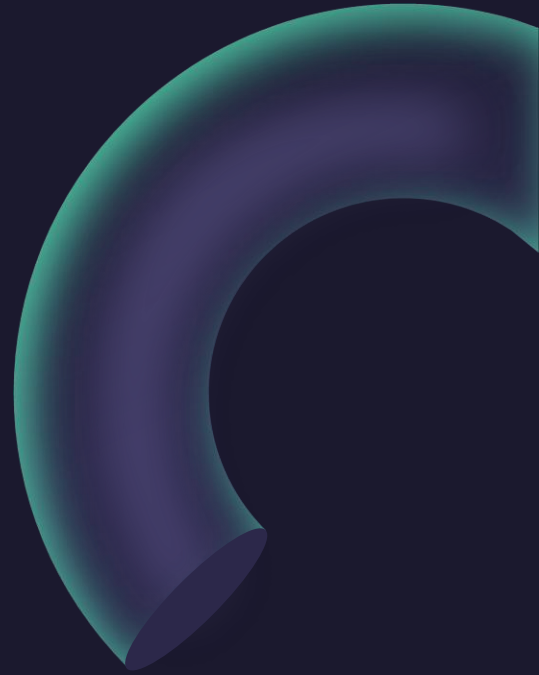
$$\int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

Line Integrals

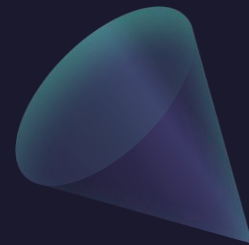
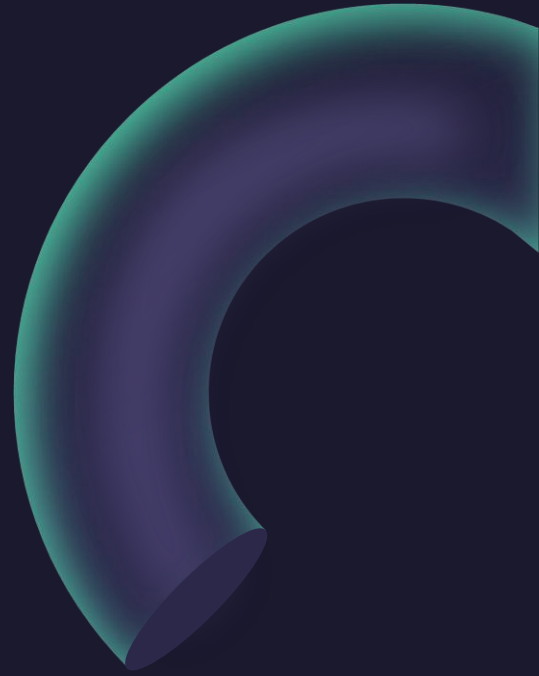
$$\int_C f(x, y, z) \, ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \, dt$$

$$\int_C P(x, y, z) \, dx + Q(x, y, z) \, dy + R(x, y, z) \, dz$$

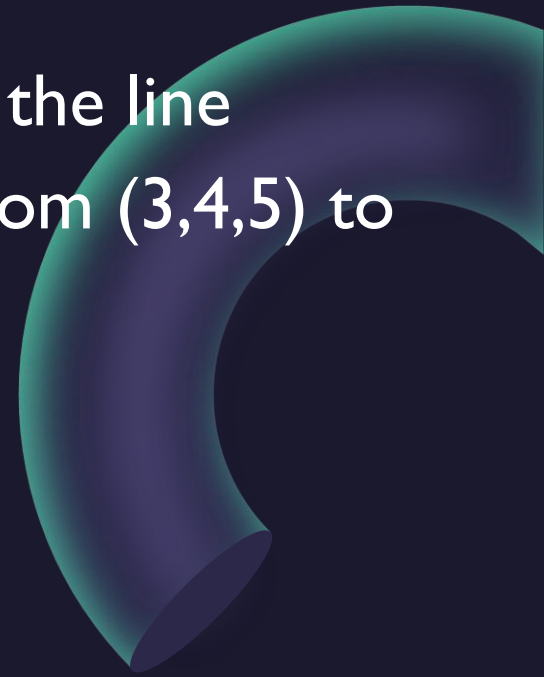
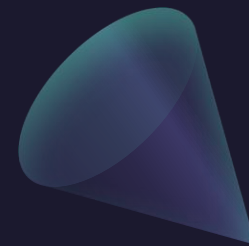
Example: Evaluate $\int_C y \sin(z) ds$ where C is the helix given by $(\cos(t), \sin(t), t)$, $0 \leq t \leq 2\pi$



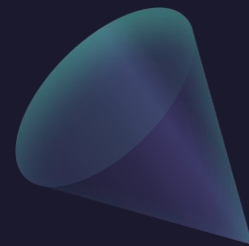
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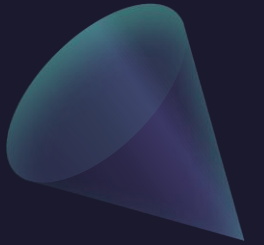
Example: Evaluate $\int_C y \, dx + z \, dy + x \, dz$ where C consists of the line segment from $(2,0,0)$ to $(3,4,5)$, followed by the line segment from $(3,4,5)$ to $(3,4,0)$.



Example: Evaluate $\int_C y \, dx + z \, dy + x \, dz$ where C consists of the line segment from $(2,0,0)$ to $(3,4,5)$, followed by the line segment from $(3,4,5)$ to $(3,4,0)$.

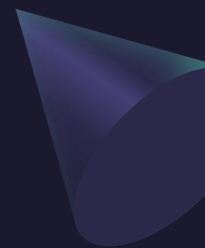


Questions?





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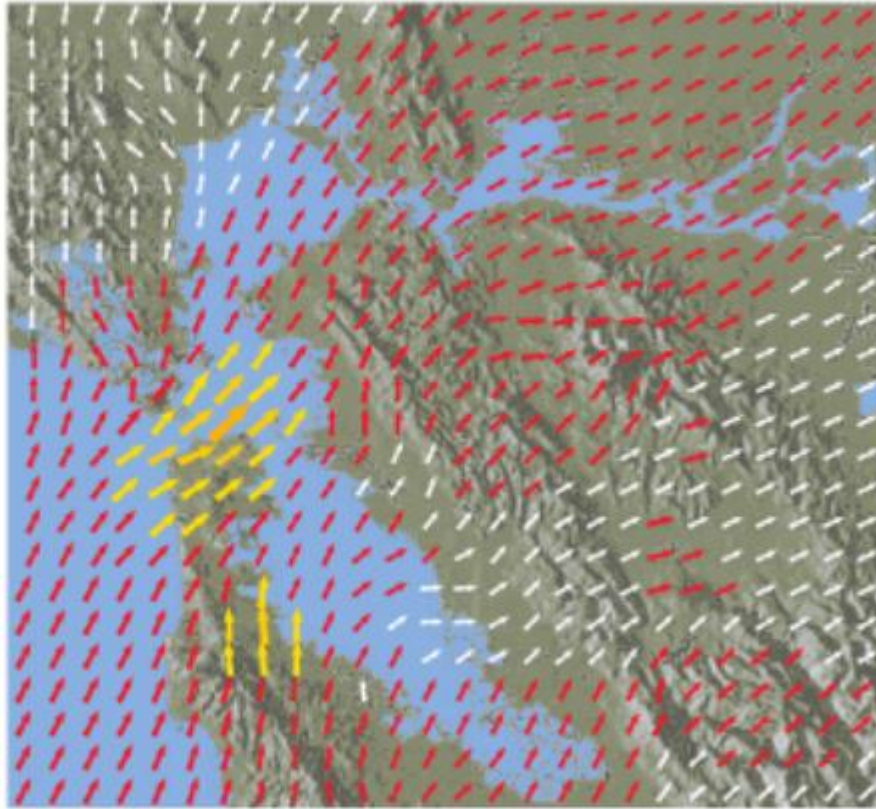
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Vector Fields

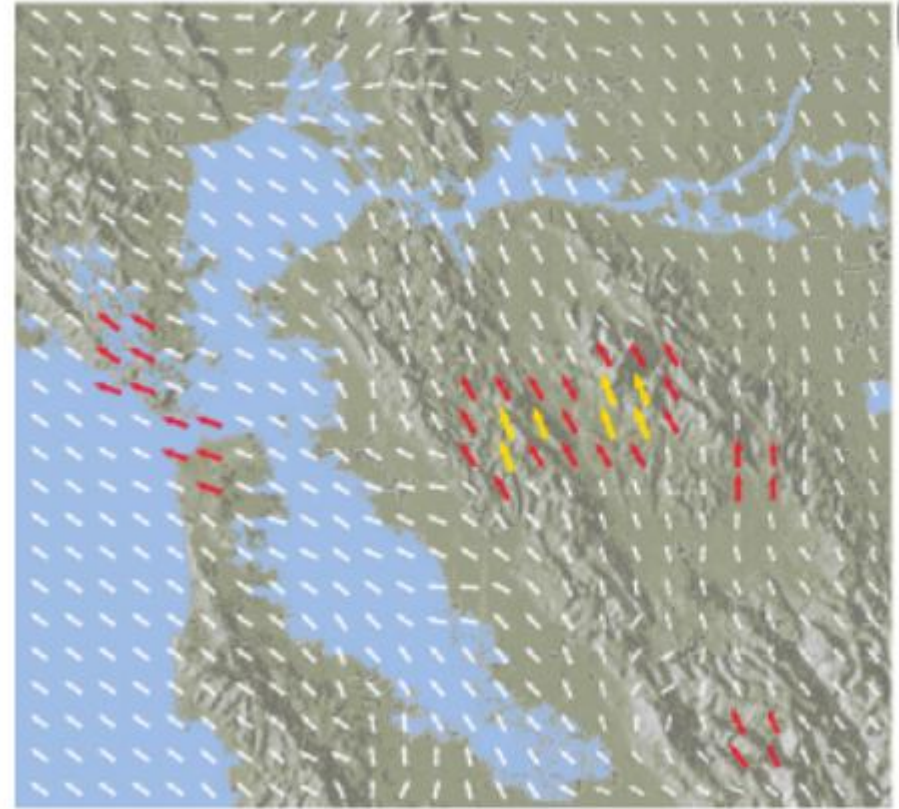


Vector Fields

Velocity vector fields showing San Francisco Bay wind patterns on a particular spring day



(a) 6:00 PM



(b) 6:00 AM

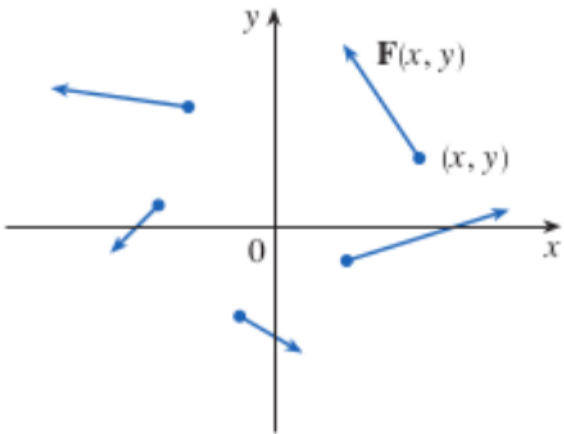
Vector Fields



Vector Fields

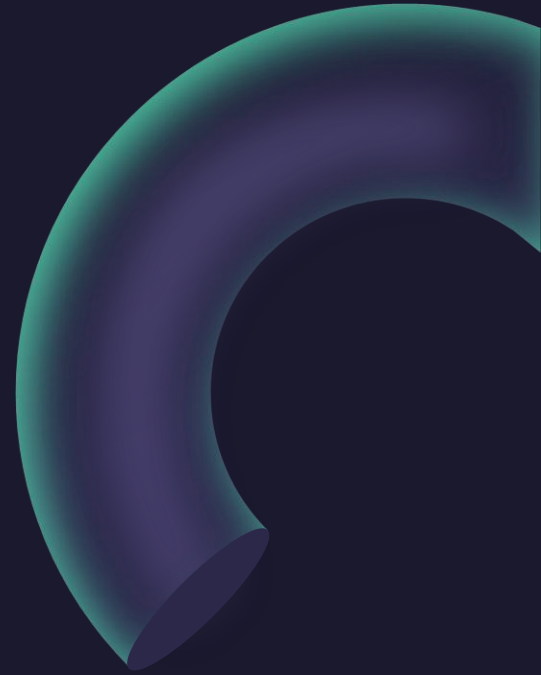
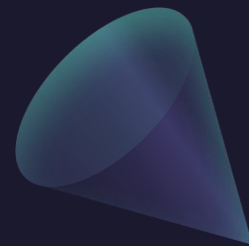
Let D be a set in $\mathbb{R}^2$ (a plane region). A **vector field on $\mathbb{R}^2$** is a function $\mathbf{F}$ that assigns to each point (x, y) in D a two-dimensional vector $\mathbf{F}(x, y)$.

Let E be a subset of $\mathbb{R}^3$. A **vector field on $\mathbb{R}^3$** is a function $\mathbf{F}$ that assigns to each point (x, y, z) in E a three-dimensional vector $\mathbf{F}(x, y, z)$.

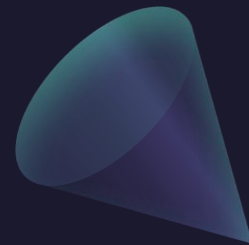
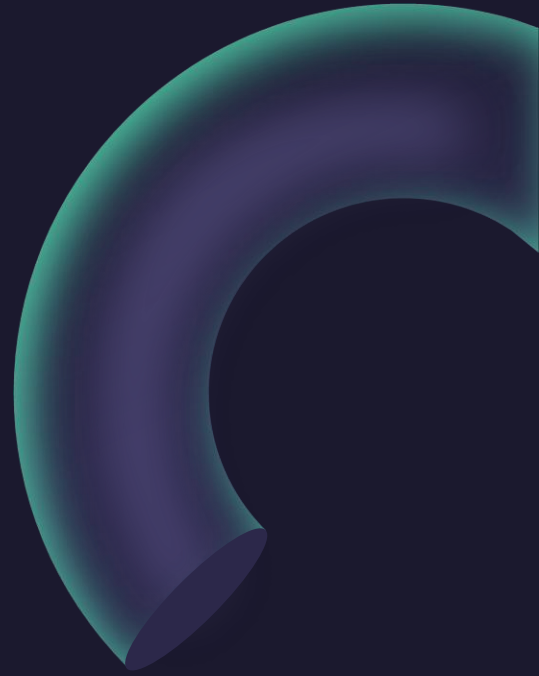


Example: Sketch a vector field on the plane given by

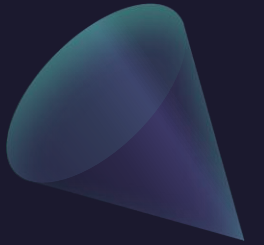
$$F(x, y) = (-y, x) = -y \cdot i + x \cdot j$$



Example: Evaluate $\int_C y \sin(z) ds$ where C is the helix given by $(\cos(t), \sin(t), t)$, $0 \leq t \leq 2\pi$



Questions?



Thank you

Until next time.

